



Qualification Pack

Engineer - Power Distribution

QP Code: PSS/Q2103

Version: 2.0

NSQF Level: 6

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PSS/Q2103: Engineer - Power Distribution

Brief Job Description

The incumbent in the job is responsible for power distribution system's erection and commissioning, operation & maintenance, testing & inspection of LT and HV systems and metering, billing, collection for distribution.

Personal Attributes

The job requires the individual should have to work outdoors in all weather conditions. The individual should have the ability to perform engineering and co-ordination activities as and when required.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1338: Apply basic health And safety practices for power related work](#)
2. [PSS/N1336: Working effectively with others](#)
3. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
4. [PSS/N2108: supervise power distribution system operation and maintenance activities](#)
5. [PSS/N2107: Automation and IoT](#)
6. [PSS/N2109: oversee installation activities of power distribution system](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Supervisory - Distribution
Country	India
NSQF Level	6
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2151.0300



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Minimum Educational Qualification & Experience	B.E./B.Tech (Electrical/ Electronics/ Instrumentation Engineering) OR Completed 2nd year diploma after 12th (Electrical/ Electronics/ Instrumentation Engineering) with 3 Years of experience OR Completed 3 year diploma after 10th (Electrical/ Electronics/ Instrumentation Engineering) with 4.5 years of experience relevant Experience OR Previous relevant Qualification of NSQF Level (level 5) with 3 Years of experience relevant Experience OR Previous relevant Qualification of NSQF Level (level 5.5) with 1.5 years of experience relevant Experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQF Approval Date	08/05/2025
Version	2.0
Reference code on NQR	QG-06-PW-04248-2025-V2-PSSC
NQR Version	2.0



Qualification Pack

PSS/N1338: Apply basic health And safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly



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PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1338
NOS Name	Apply basic health And safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026



Qualification Pack

PSS/N2108: supervise power distribution system operation and maintenance activities

Description

This OS unit covers the operation and maintenance of distribution system like substation, distribution transformer, O/H line, U/G cable, switchgear and associated components etc. ensuring uninterrupted operation of distribution system equipment.

Scope

The scope covers the following :

- Perform operation and maintenance of distribution system
- Troubleshooting and repairing of faulty equipment

Elements and Performance Criteria

Perform operation and maintenance of distribution system

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the fundamental concepts of power system security, including normal, alert, emergency, and restorative operating
- PC2.** Describe common threats to power system security, such as contingencies (e.g., generator outages, transmission line failures) and cyber/physical attacks.
- PC3.** Outline the key functions of power system security assessment, including system monitoring, contingency analysis, and corrective action analysis.
- PC4.** Identify the basic principles of power system protection schemes and their role in maintaining system integrity during fault conditions.
- PC5.** ensure smooth operation of distribution substation, overhead lines and underground cables
- PC6.** review load management, grid stability, frequency, load dispatch, feeder loading etc.
- PC7.** plan and supervise predictive, preventive, breakdown and routine maintenance for lines and substation equipment
- PC8.** locate the conduit, cables & other underground system to perform maintenance work
- PC9.** monitor performance of critical systems such as RTU, Remote Metering Unit (RMU) and other automation system
- PC10.** create SOPs, schedules, maintenance schedules
- PC11.** prepare estimates, bill of quantity for carrying out maintenance activity
- PC12.** locate faults in the distribution system lines, cable and other equipment

Troubleshooting and repairing of faulty equipment

To be competent, the user/individual on the job must be able to:

- PC13.** supervise troubleshooting and repairing / replacement activities of faulty/ unhealthy equipment
- PC14.** upgrade or modify the existing unhealthy equipment/system
- PC15.** implement technical change in equipment/system



Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant operational and maintenance standards, policies, and procedures followed in the organization
- KU2.** how to engage with specialists for support in order to resolve incidents and service request
- KU3.** operational aspects of distribution substations and lines
- KU4.** use of tools and technologies available for conducting maintenance activities
- KU5.** load management, system stability, frequency, load dispatch, etc.
- KU6.** maintenance process of transformer, substation equipment including relay and breaker
- KU7.** live and dead line maintenance of distribution system
- KU8.** emergency restoration procedures of distribution network
- KU9.** common electricity terminology and correct interpretation of the same terminology: e.g. current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
- KU10.** ratings and specifications of cables, fuses, switches and wires

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information related to work and processes
- GS2.** write reports and observations related to work in English/regional language
- GS3.** read and interpret and process flowchart for all operations
- GS4.** read manuals and operation documents to understand the Equipment used into operation
- GS5.** discuss task lists, schedules and activities with the seniors and team members
- GS6.** follow organization rule-based decision making process
- GS7.** take decisions with systematic course of actions and/or response
- GS8.** plan and organize tasks to meet deadlines
- GS9.** find ways of modifying difficult operating stages to make it operation friendly
- GS10.** apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS11.** analyze the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist
- GS12.** recognize a workplace problem and take suitable action to resolve it

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform operation and maintenance of distribution system</i>	21	53	-	-
PC1. Explain the fundamental concepts of power system security, including normal, alert, emergency, and restorative operating	2	5	-	-
PC2. Describe common threats to power system security, such as contingencies (e.g., generator outages, transmission line failures) and cyber/physical attacks.	-	-	-	-
PC3. Outline the key functions of power system security assessment, including system monitoring, contingency analysis, and corrective action analysis.	-	-	-	-
PC4. Identify the basic principles of power system protection schemes and their role in maintaining system integrity during fault conditions.	-	-	-	-
PC5. ensure smooth operation of distribution substation, overhead lines and underground cables	2	5	-	-
PC6. review load management, grid stability, frequency, load dispatch, feeder loading etc.	3	6	-	-
PC7. plan and supervise predictive, preventive, breakdown and routine maintenance for lines and substation equipment	2	6	-	-
PC8. locate the conduit, cables & other underground system to perform maintenance work	3	6	-	-
PC9. monitor performance of critical systems such as RTU, Remote Metering Unit (RMU) and other automation system	3	7	-	-
PC10. create SOPs, schedules, maintenance schedules	2	6	-	-
PC11. prepare estimates, bill of quantity for carrying out maintenance activity	2	7	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. locate faults in the distribution system lines, cable and other equipment	2	5	-	-
<i>Troubleshooting and repairing of faulty equipment</i>	9	17	-	-
PC13. supervise troubleshooting and repairing / replacement activities of faulty/ unhealthy equipment	3	7	-	-
PC14. upgrade or modify the existing unhealthy equipment/system	3	6	-	-
PC15. implement technical change in equipment/system	3	4	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2108
NOS Name	supervise power distribution system operation and maintenance activities
Sector	Power
Sub-Sector	Distribution
Occupation	Supervisory - Distribution
NSQF Level	6
Credits	5.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

PSS/N2107: Automation and IoT

Description

This NOS unit covers installing, integrating, and supervising smart grid automation and IoT systems in power distribution. It includes SCADA, smart meters, RTUs, AMI platforms, and field sensors. The job involves interpreting network designs, ensuring device compliance, interfacing control systems, and analyzing real-time data for efficiency, reliability, and cybersecurity.

Scope

The scope covers the following :

- The role involves identifying and installing automation and IoT components for smart grids, interfacing field devices with control centres while ensuring protocol compliance, and monitoring as well as analysing smart grid performance data using IoT tools. It also includes troubleshooting system issues and coordinating with relevant stakeholders to maintain efficient and reliable grid operations.

Elements and Performance Criteria

Smart Grid Automation and IoT Integration

To be competent, the user/individual on the job must be able to:

- PC1.** identify IoT and automation technologies used in power distribution systems such as SCADA, smart meters, RTUs, and AMI platforms
- PC2.** interpret network design requirements to integrate IoT-based solutions for monitoring and control of LT/HT feeders, substations, and distribution transformers
- PC3.** coordinate installation and commissioning of smart grid components like IoT-enabled meters, fault indicators, and data loggers
- PC4.** interface field automation devices (sensors, relays, actuators) with control centres or cloud-based dashboards
- PC5.** ensure compliance of installed automation devices with communication protocols such as Modbus, DNP3, or IEC 61850
- PC6.** monitor data streams from field devices and identify anomalies, energy losses, or performance degradation using IoT analytics
- PC7.** supervise troubleshooting and rectification of issues related to connectivity, configuration, or power supply of IoT devices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Structure and functions of smart grid automation systems (SCADA, AMI, RTUs, smart meters) in power distribution
- KU2.** Communication protocols (e.g., Modbus, DNP3, IEC 61850, Zigbee, RF Mesh) and their applications



Qualification Pack

- KU3.** Integration architecture for IoT devices in LT/HT distribution networks and substations
- KU4.** Standard procedures for installing, configuring, and testing IoT-based control and monitoring systems
- KU5.** Common hardware and software issues in smart grid setups and methods for resolution
- KU6.** Data analysis techniques to interpret energy usage, fault reports, and power quality metrics from IoT platforms
- KU7.** Safety and cybersecurity considerations in automation and IoT deployment in power systems
- KU8.** Documentation, reporting tools, and protocols for coordination with DISCOMs, vendors, and project teams

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Record observations, configurations, and changes for reference
- GS2.** Read user manuals, wiring diagrams, IoT protocols, and safety guidelines
- GS3.** Explain setup to customers and team report technical issues
- GS4.** Understand customer concerns, feedback, and technical requirements
- GS5.** Decide on best protocol/device compatibility and troubleshoot issues
- GS6.** Sequence installation and configuration tasks efficiently
- GS7.** Resolve installation, connectivity, and integration errors
- GS8.** Assess signal strength, latency, and system health
- GS9.** Demonstrate features and assist customers in IoT system usage
- GS10.** Follow installation and circuit modification safety standards

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Smart Grid Automation and IoT Integration</i>	30	70	-	-
PC1. identify IoT and automation technologies used in power distribution systems such as SCADA, smart meters, RTUs, and AMI platforms	5	12	-	-
PC2. interpret network design requirements to integrate IoT-based solutions for monitoring and control of LT/HT feeders, substations, and distribution transformers	4	10	-	-
PC3. coordinate installation and commissioning of smart grid components like IoT-enabled meters, fault indicators, and data loggers	5	12	-	-
PC4. interface field automation devices (sensors, relays, actuators) with control centres or cloud-based dashboards	5	12	-	-
PC5. ensure compliance of installed automation devices with communication protocols such as Modbus, DNP3, or IEC 61850	5	12	-	-
PC6. monitor data streams from field devices and identify anomalies, energy losses, or performance degradation using IoT analytics	3	6	-	-
PC7. supervise troubleshooting and rectification of issues related to connectivity, configuration, or power supply of IoT devices	3	6	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2107
NOS Name	Automation and IoT
Sector	Power
Sub-Sector	
Occupation	Supervisory - Distribution
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

PSS/N2109: oversee installation activities of power distribution system

Description

This OS unit is about overlooking all installation related activities including erection and commissioning of power distribution systems.

Scope

The scope covers the following :

- Installation of power distribution system
- Erection and commissioning of substation

Elements and Performance Criteria

Installation of power distribution system

To be competent, the user/individual on the job must be able to:

- PC1.** interpret schematic drawings, engineering drawings, single line diagrams, lay out plans etc. of the power distribution system to identify the installation requirements
- PC2.** supervise route survey for O/H line, U/G lines and ROW, for installation of new service line connections
- PC3.** manage revenue across all processes in distribution chain from release of new connection to collection including meter installation, meter reading, bill generation, bill distribution etc.
- PC4.** prepare revenue estimates and bill of quantities (BoQ)
- PC5.** overview functioning of other departments such as procurement (modes and types- turnkey, supply & services etc.), material planning and handling, store handling, logistics etc
- PC6.** supervise installation of various power system equipment such as transformers, circuit breakers, isolators, current transformers, voltage transformers, wave traps, etc.
- PC7.** review operation of all types of protection system and earthing in distribution network
- PC8.** implement new technologies such as Smart grid, AMR, AMI and SCADA and GIS mapping in the substation

Erection and commissioning of substation

To be competent, the user/individual on the job must be able to:

- PC9.** supervise erection and commissioning of substation equipment and line elements like different types of poles/ structures , O/H line, U/G cable, switchgear, control panel etc.
- PC10.** supervise installation of Substation Automation System (SAS), pole mounting substation, earthing, G.O. switch etc. and protection system such as circuit breaker, relay, CT, PT and LA in the substation
- PC11.** ensure that all distribution system equipment are protected and have proper grounding and earthing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** relevant legislation, standards, policies, and procedures followed in the organization
- KU2.** how to engage with specialists for support in order to resolve incidents and service requests
- KU3.** how to read network layout, schematic and design drawing of substation
- KU4.** the air insulated and gas insulated substation
- KU5.** technical specification of distribution system equipment
- KU6.** design of distribution network based on prevailing planning and policy guidelines
- KU7.** types of distribution system network - ring main system, radial system, interconnected system etc.
- KU8.** different types of transformers, poles, conductors and cables
- KU9.** load flow studies, sag and tension calculation, transient studies, vibration analysis, wind pressure analysis, short circuit studies etc.
- KU10.** smart grid, AMR, AMI, automation system and various types of protection equipment
- KU11.** ratings and specifications of line, transformer, cables, fuses, switches and wires
- KU12.** how to handle and use all machineries, equipment's & vehicles
- KU13.** how to estimate time, material and equipment needed to complete assignments
- KU14.** procedures for storing raw materials and finished goods

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information related to work and processes
- GS2.** write reports and observations related to work in English/regional language
- GS3.** read and interpret and process flowchart for all operations
- GS4.** read manuals and operation documents to understand the equipment used into operation
- GS5.** discuss task lists, schedules and activities with the seniors and team members
- GS6.** follow organization rule-based decision making process
- GS7.** take decisions with systematic course of actions and/or response
- GS8.** plan and organize tasks to meet deadlines
- GS9.** find ways of modifying difficult operating stages to make it operation friendly
- GS10.** apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS11.** analyze the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist
- GS12.** report problems and resolve in a timely manner

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation of power distribution system</i>	21	52	-	-
PC1. interpret schematic drawings, engineering drawings, single line diagrams, lay out plans etc. of the power distribution system to identify the installation requirements	2	5	-	-
PC2. supervise route survey for O/H line, U/G lines and ROW, for installation of new service line connections	2	5	-	-
PC3. manage revenue across all processes in distribution chain from release of new connection to collection including meter installation, meter reading, bill generation, bill distribution etc.	2	5	-	-
PC4. prepare revenue estimates and bill of quantities (BoQ)	2	5	-	-
PC5. overview functioning of other departments such as procurement (modes and types- turnkey, supply & services etc.), material planning and handling, store handling, logistics etc	4	10	-	-
PC6. supervise installation of various power system equipment such as transformers, circuit breakers, isolators, current transformers, voltage transformers, wave traps, etc.	4	11	-	-
PC7. review operation of all types of protection system and earthing in distribution network	2	5	-	-
PC8. implement new technologies such as Smart grid, AMR, AMI and SCADA and GIS mapping in the substation	3	6	-	-
<i>Erection and commissioning of substation</i>	9	18	-	-
PC9. supervise erection and commissioning of substation equipment and line elements like different types of poles/ structures , O/H line, U/G cable, switchgear, control panel etc.	3	6	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. supervise installation of Substation Automation System (SAS), pole mounting substation, earthing, G.O. switch etc. and protection system such as circuit breaker, relay, CT, PT and LA in the substation	3	6	-	-
PC11. ensure that all distribution system equipment are protected and have proper grounding and earthing	3	6	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2109
NOS Name	oversee installation activities of power distribution system
Sector	Power
Sub-Sector	Distribution
Occupation	Supervisory - Distribution
NSQF Level	6
Credits	6
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Guidelines will be same as that of QP

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1338.Apply basic health And safety practices for power related work	39	61	-	-	100	10
PSS/N1336.Working effectively with others	34	56	-	-	90	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
PSS/N2108.supervise power distribution system operation and maintenance activities	30	70	-	-	100	30
PSS/N2107.Automation and IoT	30	70	-	-	100	20
PSS/N2109.oversee installation activities of power distribution system	30	70	-	-	100	20
Total	183	357	-	-	540	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.